Health Spending and Life Expectancy Across Economies in 2019

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Repository: https://github.com/Zhengdao-Lu/Midterm-Project-Econometric-Foundation

Notebook: https://colab.research.google.com/drive/110U0DoUVaKBcL1ssJKVsYfZt2drbkgvk

Question	How are health spending per person and urbanization related to life expectancy at birth across economies in 2019, after accounting for GDP per person?
Headline finding	In the 186 complete cases, doubling health spending per person is associated with 1.35 more years of life expectancy after accounting for GDP and urbanization
Data	Four World Bank World Development Indicators series for 2019. One row is one country or economy; the regression uses 186 complete cases.
Method	OLS of life expectancy on log health spending, urban population share, and log GDP per person, with HC1 robust standard errors.

1. The Question and the Data

Nowadays, countries spend very different amounts on health care, and people don't have the same life expectancy across countries. My question is how per-person health spending and urbanization are related to life expectancy at birth across countries and economies in 2019, after accounting for GDP per person. This question matters to researchers and policymakers who compare health outcomes across economies. I expect higher spending to go with longer life expectancy, but richer economies can both spend more and have other advantages that help people live longer. That is why I include GDP before interpreting the relationship between spending and life expectancy.

The data used in the study come from four World Bank World Development Indicators (WDI) files. The World Bank compiles the figures, but the underlying data come from different sources. Life expectancy draws on UN and national mortality statistics. The WHO supplies the health spending series. GDP per person comes from national accounts and purchasing power estimates, while the urban population share uses UN estimates and national definitions of urban areas. Using one database keeps the country codes and years consistent, but it does not make every measure equally precise.

Each variable measures a particular part of a broader idea. Life expectancy at birth describes how long a newborn would live if mortality rates during that period remained the same. Current health expenditure includes spending from several financing sources, so it measures more than a government health budget. It records spending, but does not tell me whether people received effective care. The urban share also needs care because countries do not all define an urban area the same way. With those limits in mind, I focus on a single year before the pandemic.

I use 2019 because it predates the COVID-19 pandemic, which affected life expectancy in many places. Life expectancy is measured in years, and urbanization is expressed as a percentage.

Health spending is in current PPP international dollars per person, and GDP per person is in constant 2021 PPP international dollars. PPP adjusts for price differences across economies. The word "2021" in the GDP unit names the price base, not the year of the observation; every value in this report is from 2019. The two dollar series use different price bases, so I analyze them separately and do not divide one by the other.

The four original files each contain 266 rows, including regional totals, income groups, and other aggregate or unclassified entries. I keep the 2019 column, remove 49 such entries from each file, and join the remaining observations by country code. The merged data have 217 rows and 7 columns: country name, country code, year, and the four measures. One row is one country or economy in 2019. This is a one-year comparison, so the data cannot show whether a change in spending came before a change in life expectancy.

Missing data reduce the usable sample. GDP per person is missing for 18 economies and health spending for 24, while life expectancy and urbanization have no gaps in the merged table. Some gaps overlap, leaving 186 economies with all four measures. For example, GDP and health spending are both absent for American Samoa and Guam; GDP is absent for South Sudan and Yemen, while health spending is absent for Bermuda and the Cayman Islands. These examples show that the gaps are concentrated in particular kinds of economies, so I do not call them missing completely at random (MCAR). Missing at random (MAR) after accounting for observed characteristics is possible, but the files cannot rule out missing not at random (MNAR) if reporting also depends on the unobserved values.

I use complete cases rather than filling in gaps with averages. That choice keeps the figures and model based on the same set of 186 economies, although the results may describe economies with missing information less well. Table 1 reports the original units for this analysis sample. Life expectancy averages 72.28 years, with a median of 73.01. The lowest value, 31.53 years, is for the Central African Republic in the downloaded file. I keep it for the main analysis and check its influence in the diagnostics.

Table 1. Summary statistics

Variable (units)	Mean	Median	SD	Min	Max	N
Life Expectancy (Years)	72.28	73.01	8.03	31.53	85.26	186
Health Spending (Current PPP $/person)	1610.30	821.59	1929.51	36.11	10550.22	186
Urban population(%)	59.60	61.18	22.33	13.93	100	186
GDP (Constant 2021 PPP $/person)	24148.58	15390.42	24459.40	1054.73	132724.75	186

Source: World Bank World Development Indicators, 2019. The table uses the 186 economies with all four variables.

2. Descriptive Evidence

Table 1 shows large differences in spending and income before any model is fitted. Median health spending is 821.59 current PPP dollars per person, while its mean is 1,610.30. Median GDP per person is 15,390.42 constant 2021 PPP dollars, compared with a mean of 24,148.58. In both cases, a smaller group of high values pulls the mean above the median. Health spending ranges from 36.11 to 10,550.22, and GDP ranges from 1,054.73 to 132,724.75 in their respective PPP units. These wide ranges make the distributions important to see before interpreting a single regression coefficient.

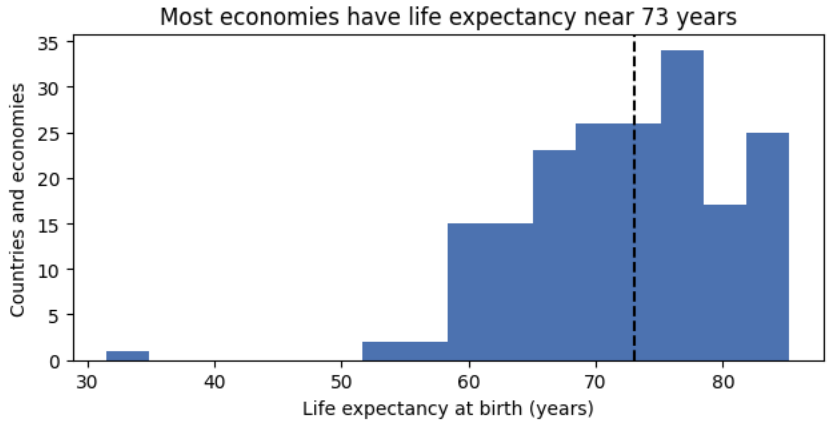


Figure 1. Most economies have life expectancy near 73 years, while only a small number are lower. Source: World Bank life expectancy series, 2019.

Figure 1 shows that most economies have life expectancy near the median of 73.01 years. The lower tail includes the Central African Republic, whose value is far below the rest of the sample. I leave life expectancy in years because a difference in years is easy to interpret. The unusually low value may influence the regression, so I will return to it using Cook's distance later. For the spending comparison, the long upper tail is a more immediate problem: values near the low end are hard to distinguish on a dollar scale.

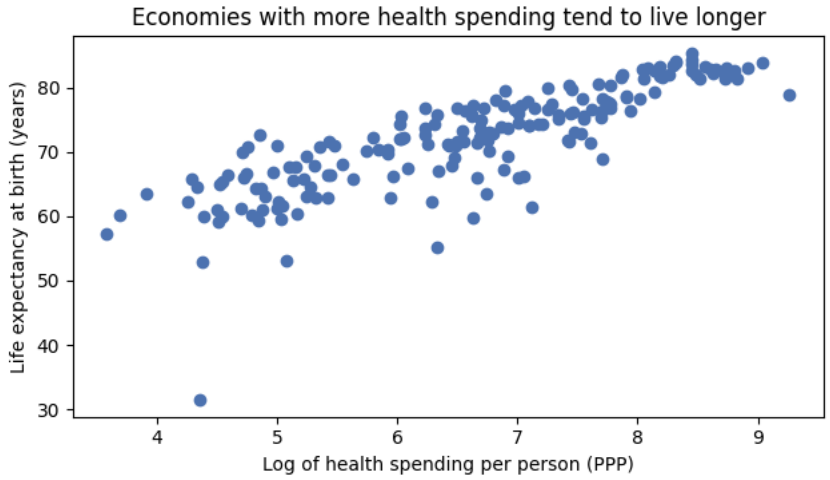


Figure 2. Economies that spend more per person on health generally have higher life expectancy, even after accounting for GDP and urbanization. Source: World Bank WDI, 2019.

Figure 2 plots life expectancy against the natural log of health spending per person. The points generally rise from left to right, which agrees with the positive relationship I expected. Health spending has a right-skewness of 1.83 in the original units, and GDP has a right-skewness of 1.71. Logging these variables spreads out the lower values and compresses the long upper tails. The scatterplot still shows an unadjusted relationship: richer economies may spend more on health and have longer life expectancy for reasons beyond spending.

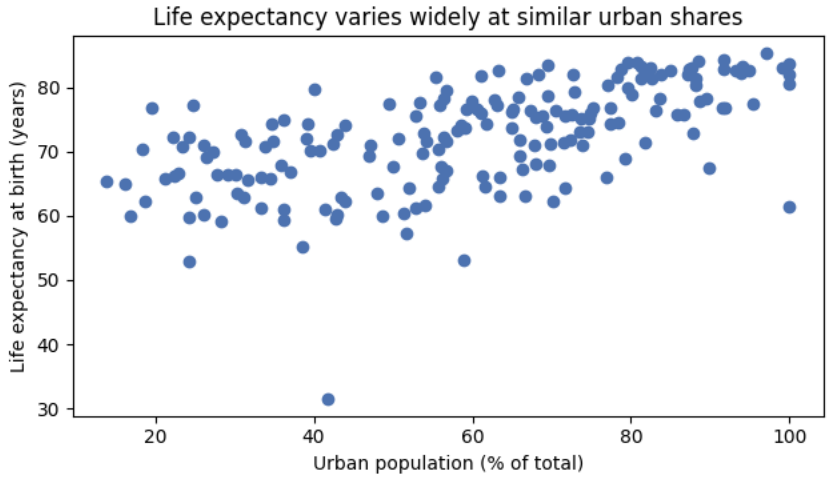


Figure 3. Life expectancy differs substantially among economies with similar urban population shares. Source: World Bank WDI, 2019.

Figure 3 shows a less orderly pattern of urbanization: economies with similar urban shares can have very different life expectancy levels. Thus, urbanization itself does not capture the difference and should be considered alongside health spending and GDP in the regression. The share says how many people live in areas classified as urban, but it does not measure sanitation, housing quality, or access to care within those areas. All three figures use the same complete-case sample as the model. The life expectancy axes on the scatterplots start near the lowest observed value, making differences among economies easier to see. The histogram also includes the unusually low observation. Each economy appears as one point, regardless of its population size.

3. The Model

I use an ordinary least squares (OLS) model, with life expectancy at birth (in years) as the outcome. The variables are log health spending, urban population share, and log GDP per person. OLS chooses the line that makes the sum of squared differences between observed and predicted life expectancy as small as possible. The predictors are the natural logs of health spending and GDP per person, along with the urban population percentage. For country i, the model can be written as follows:

$$\text{Life expectancy}_i = \beta_0 + \beta_1 \log(\text{health spending}_i) + \beta_2 \text{ urban share}_i + \beta_3 \log(\text{GDP}_i) + u_i$$

GDP per capita is included in the model, as national income will obviously affect both health spending and life expectancy. Higher-income countries will be able to afford more health care and will presumably also have better living conditions. Including GDP allows us to look at how

spending varies with life expectancy among countries at similar levels of income. It doesn't take into account other differences (e.g., sanitation, education). GDP could also be a poor control for a long-run causal question if earlier health spending had in fact been the thing that raised GDP. That possibility is not ruled out by this one-year analysis, so I treat the coefficient as an association.

I use natural logs for the two monetary variables because a fixed dollar difference means something different in a low-spending economy than in a high-spending one. A logged predictor lets the coefficient describe proportional differences. I explain the health result as a doubling of spending, and the urban result as a 10-percentage-point difference in the share of the population living in urban areas. These changes in wording come from the fitted coefficients; they do not change the model.

The model uses 186 countries and has an R-squared of 0.727. Table A1 presents the coefficients and HC1-robust standard errors. The coefficient on log health spending is 1.950. Thus, the model shows that, when holding GDP and urbanization constant, a one-unit increase in the log of spending corresponds to 1.950 additional years of life expectancy. Doubling per-person health spending leads to 1.35 more years of life expectancy. The robust standard error is 0.905, and the translated 95% confidence interval for doubling is 0.12 to 2.58 additional years. Both ends are positive, but the range is fairly wide, making the exact magnitude of the association uncertain.

Urbanization is measured in percentage points. With health spending and GDP at the same values, a ten-percentage-point higher urban share corresponds to 0.02 fewer years of life expectancy. The 95% confidence interval runs from 0.45 fewer years to 0.41 more years. It includes zero and small differences in either direction, so the data do not show a clear relationship with urbanization once the other variables are included in the model. This is different from saying that urban living has no effect; the model estimates an economy-level association in one year.

An R-squared of 0.727 means the three predictors together account for about 72.7% of the variation in life expectancy among the 186 economies in the sample. It does not show that any one coefficient is causal. The positive GDP coefficient in Table A1 is also conditional on the other two predictors. Because spending and GDP move closely together, the model provides limited information for precisely separating their individual relationships. The diagnostics help show how much uncertainty remains in the estimates.

4. Diagnostics

Figure A1 plots the residuals against fitted life expectancy and against log health spending. A residual is the observed life expectancy minus the value predicted by the model. The plots show an uneven spread around zero and one large negative residual. The negative point is the Central African Republic: its observed life expectancy is much lower than the model predicts from its spending, GDP, and urban share. The plots do not explain why that happened, but they show where the simple model fits poorly.

To check whether the residual variance changes with the model's predictors, the breusch-pagan test was taken. The p-value is 0.0073, indicating that the spread is uneven. Therefore, Table A1 reports the HC1-robust standard errors. It changes the estimated uncertainty, confidence intervals, and the p-values. However, it will not affect the OLS coefficients. It addresses the uneven residual spread, but it cannot correct for an omitted variable or establish cause and effect. To assess how much a predictor overlaps with other predictors in the model, variance inflation factors (VIFs) were calculated. They are 9.09 for log health spending, 8.97 for log GDP, and 1.99 for urban share. The two high values show that spending and GDP contain much of the same cross-economy information. As a result, it is harder to estimate their separate coefficients precisely. Urbanization has less overlap with the other predictors. I keep GDP in the model because it is a plausible confounder, while being cautious about the exact magnitude of the spending coefficient.

In addition, I also check whether one economy has too much influence on the fitted line. Based on the Cook's distance for the countries, the Central African Republic has the largest value, 0.481, exceeding the 4/N screening line of 0.022. Its life expectancy is 31.53 years, as reported in the source file. When I leave this economy out and fit the same model again, the health coefficient rises from 1.950 to 2.311. The coefficient remains positive, although its size changes. I keep the economy in the main model because Cook's distance identifies influence, not an error. Deleting an observation simply because it changes a result would hide a real sensitivity.

5. Threats and Conclusion

The first limitation on interpreting the spending coefficient is omitted-variable bias. The model does not measure education, sanitation, or health-system quality. Some of these conditions may be better in economies that spend more on health, and they can also raise life expectancy. If so, part of the positive spending coefficient would reflect those conditions. For example, if better sanitation is common in higher-spending economies, omitting it could make the estimated spending relationship appear stronger than it would be after sanitation is measured.

The second limitation is Reverse causality. Greater spending might help people live longer, but a longer-lived population may also need more health care and spend more. Since spending and life expectancy are both measured in 2019, these data cannot show which change came first. The positive coefficient therefore cannot be read as proof that increasing a health budget would add the estimated number of years to life expectancy.

The third limitation is the missing data. The regression excludes counties without GDP or health spending data, such as several small territories. Thus, if the spending relationship differs across those countries, the estimation based on the 186 countries cannot fully represent the results for the merged data of the 217 countries. The dataset shows which values are absent, but not what the missing values would have been; Therefore, I'm unable to determine whether including them would raise or lower the coefficient.

In conclusion, higher per-person health spending is associated with longer life expectancy among the countries studied in 2019. In the OLS model, which includes GDP per person and urban share, doubling spending is associated with an increase of 1.35 additional years, with a 95%

confidence interval of 0.12 to 2.58 years. The diagnostic checks support the direction of this result. The interval remains above zero, and the spending coefficient remains positive when the Central African Republic, the country with the highest Cook's distance, is excluded. Its exact size remains uncertain because the interval is wide and that economy changes the coefficient. Moreover, Urbanization has no clear relationship with life expectancy in this model. These results describe differences between countries, but they cannot tell us whether increasing health spending would raise life expectancy.

Work Cited:

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Appendix (does not count toward the 6 to 8 pages)

Table A1. Life expectancy regressed on health spending, urbanization, and GDP per capita

Predictor	Coefficient	HC1 SE	95% CI low	95% CI high
Intercept	22.653	6.435	10.040	35.266
Log health spending	1.950	0.905	0.176	3.725
Urban population (%)	-0.002	0.022	-0.045	0.041
Log GDP per person	3.856	1.242	1.422	6.290
N = 186				
R^2 = 0.727				

Dependent variable: life expectancy at birth, in years. Source: World Bank WDI, 2019. Standard errors are HC1 robust

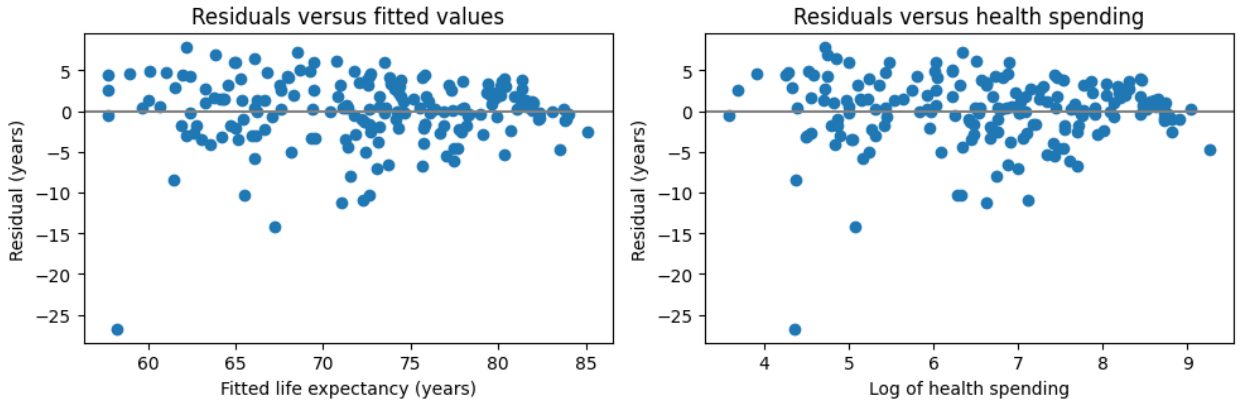


Figure A1. Residuals show uneven spread and one large negative value against fitted life expectancy and log health spending. Source: OLS model using World Bank WDI, 2019.